Triangle Geometry

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January 6, 2023

Preface

This handout offers an introductory guide to triangle geometry, including the essential basics of area formulae, similarity and congruence, trigonometry, angle bisector theorem, and Law of Sines/Cosines. Most of the featured problems have their origins in the American Mathematics Competition. There are a few more arcane topics such as Stewart's Theorem and Ceva's Theorem that don't surface in AMC 10 that often but can be crucial for other popular contests such as state math leagues and ARML. Solutions to a few selected problems are provided near the back of the guide.

Preliminary Ideas

A triangle is a polygon with three sides. We will denote its vertices as A, B, and C, and the side lengths as a, b, and c. We say that an altitude or height is a line segment that goes from a vertex to the opposing side in a perpendicular/orthogonal manner. The area of a triangle is simply $\frac{1}{2}bh$, where b is the length of a base and h is the length of the corresponding height.

More generally, the area of a triangle, given the side lengths, is

$$A = \sqrt{s(s-a)(s-b)(s-c)},$$

where s is half of the perimeter. This is called **Heron's formula**. Additionally, the perimeter is equal to $a + b + c$. We can categorize and classify triangles based on properties relating to the side lengths and angles. If one of the angles measures 90 and the longest side length is c, then $a^2 + b^2 = c^2$ via **Pythagorean Theorem**.

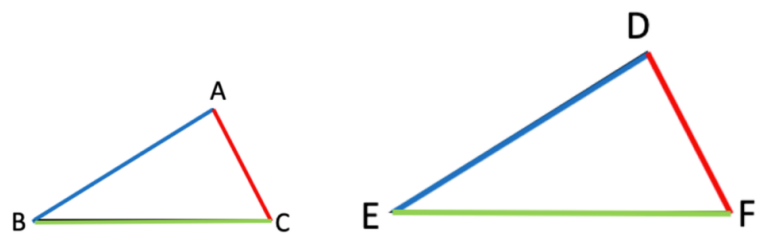
Not all tuples of side lengths a, b, and c can form a legitimate triangle. All side lengths must obey the **triangle inequality**, which colloquially states that no one side length can exceed or be equal to the sum of the other two side lengths. In other words, $a + b > c$, $a + c > b$, and $b + c > a$. As a corollary, if we let $c = \max(a, b, c)$, then ΔABC is acute if $a^2 + b^2 > c^2$, right if $a^2 + b^2 = c^2$, and obtuse if $a^2 + b^2 < c^2$.

Similar and Congruent Triangles

We saw that two triangles ΔABC AND ΔDEF are similar—notationally, $\Delta ABC \sim \Delta DEF$—if they have an identical of sets. If $\angle A = \angle D$, $\angle B = \angle E$, and $\angle C = \angle F$, then the following ratio comparison of side lengths is valid.

$$\frac{|AB|}{|DE|} = \frac{|BC|}{|EF|} = \frac{|AC|}{|DF|} = r.$$

An important note is that we can relate the **areas of two similar triangles** easily. Given the ratios above, we can prove that $\frac{[\Delta ABC]}{\Delta DEF} = r^2$ using Heron's Formula. We can investigate triangle similarity in myriad ways. If two triangles have a pair of equal angles and the two sides surrounding the angle have an equal ratio, then the two triangles are *similar*. Similarity can also be achieved if three sides of one triangle can be multiplied/divided by some k to produce the other triangle.



When two triangles have the same angles as well as the same set of side lengths, we denote the triangles as **congruent**. In terms of notation, we write $\Delta ABC \cong \Delta DEF$. To prove congruence, we can use SSS (all sides are respectively equal), SAS (two sides and the angle between them are respectively equal), AAS (two angles and any side are respectively equal), and ASA (two angles and the side between them are equal). Any approach is sufficient.

Triangle similarity and congruence are quite nice because they enable us to interrelate triangles and reduce the search space for variables and angle measures. They also allow us to forge equations for side lengths that can be crucial keys in geometric solutions.

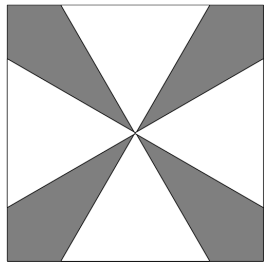
Problem Batch A

1. How many equilateral triangles of side length 1 can fit inside an equilateral triangle of side length 4?

2. A triangle has side lengths 10, 10, and 12. A rectangle has width 4 and area equal to the area of the triangle. What is the perimeter of this rectangle? (AMC 10B 2010)

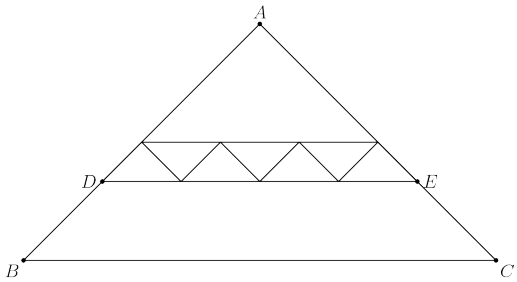
3. The two legs of a right triangle, which are altitudes, have lengths $2\sqrt{3}$ and 6. How long is the third altitude of the triangle? (AMC 10A 2014)

4. The figure below shows a square and four equilateral triangles, with each triangle having a side lying on a side of the square, such that each triangle has side length 2 and the third vertices of the triangles meet at the center of the square. The region inside the square



but outside the triangles is shaded. What is the area of the shaded region? (AMC 10B 2019)

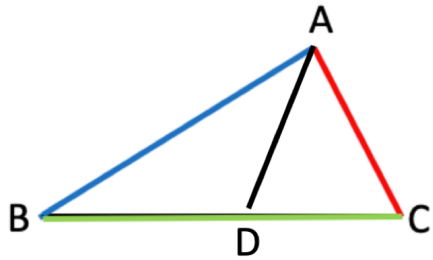
5. All of the triangles in the diagram below are similar to isosceles triangle ABC, in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$? (AMC 10A 2018)



6. Right $\triangle ABC$ has $AB = 3, BC = 4$, and $AC = 5$. Square $XYZW$ is inscribed in $\triangle ABC$ with X and Y on $\overline{AC}, W$ on $\overline{AB}$, and Z on $\overline{BC}$. What is the side length of the square? (AMC 10B 2007)

Angle Bisector Theorem

Suppose that in triangle ΔABC, we have angle bisector AD where point D lies on BC and AD splits the angle $\angle BAC$ in half such that $\angle BAD = \angle CAD$.



The following equation regarding side lengths is true.

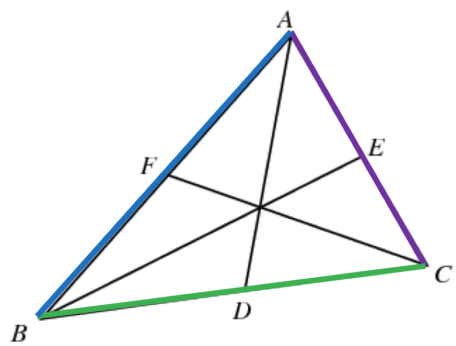
$$\frac{|AB|}{|BD|} = \frac{|AC|}{|DC|}.$$

If we have angle trisectors AD and AE with D and E on side BC such that $\angle BAD = \angle DAE = \angle EAC$, then we can separately consider the sub-triangles ΔBAE and ΔDAC and apply angle bisector theorem on each of them.

Ceva's Theorem

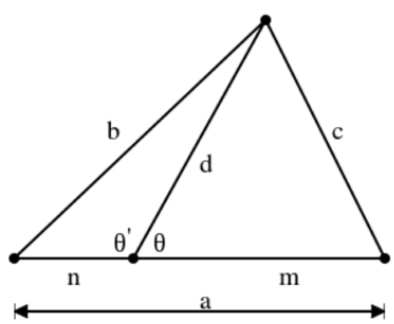
Let ΔABC be a triangle and consider cevians AD, BE, and CF with points D, E, and F on sides BC, AC, and AB respectively. A a technical note, a cevian is simply a line segment from a vertex to a point on the opposite side. According to Ceva's Theorem,

$$\frac{AE}{EC} * \frac{CD}{DB} * \frac{BF}{FA} = 1.$$



Stewart's Theorem

Let ΔABC is a triangle and point D be a point on side BC. We label the following side lengths as enumerated: $|AC| = b$, $|BC| = a$, $|AB| = c$, $|AD| = d$, $|BD| = m$, and finally $|DB| = m$. The side lengths obey the mathematical relationship $man + dad = bmb + cnc$ or equivalently $a(mn + d^2) = b^2m + c^2n$, which we can remember using the mnemonic device *"A man and his dad put a bomb in the sink"*. A proof of this theorem leverages the law of cosines on θ and θ'.



$$man + dad = bmb + cnc$$

"A man and his dad put a bomb in the sink"

$$a(mn + d^2) = b^2m + c^2n$$

Problem Batch B

1. In ΔABC, the angle bisector AI divides the median BM into two segments of length 200 and 300, and AI divides BC into two segments of length 660 and x. Find the largest possible value of x. (ARML 1980)

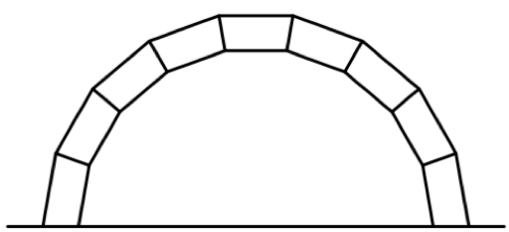
2. In ΔABC, point D is on $\overline{BC}$ such that $\overline{AD}$ bisects angle $\angle A$. If $AD = BD = AB - 1 = 5$, determine exactly the value of the perimeter of ΔABC. (MN Math League 17/18)

3. Triangle ABC has a right angle at B, $AB = 1$, and $BC = 2$. The bisector of $\angle BAC$ meets $\overline{BC}$ at D. What is BD? (AMC 10B 2009)

4. In $\triangle ABC$, $AB = 86$, and $AC = 97$. A circle with center A and radius AB intersects $\overline{BC}$ at points B and X. Moreover $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC? (AMC 10A 2013)

5. In ΔABC, $\overline{AD}$ and $\overline{AE}$ form the angle trisectors of angle $\angle A$ (they divide $\angle A$ into three angles endowed with equal measure), with D closer to point B than E. Suppose that $AB = AC = 4$ and $AD = AE = 3$. Compute the length of BC.

6. The keystone arch is an ancient architectural feature. It is composed of congruent isosceles trapezoids fitted together along the non-parallel sides, as shown. The bottom sides of the two end trapezoids are horizontal. In an arch made with 9 trapezoids, let x be the angle measure in degrees of the larger interior angle of the trapezoid. What is x?



Circles in Triangles

There are a number of circles that are intricately related to triangles. We say that the **circumcircle** of any given triangle is the circle that passes through all three of its vertices. We note that every triangle has a distinct circumcircle since any three non-collinear points determines a unique circle. For a triangle with side lengths a, b, and c, and a circumradius of R and area of A, we have the following equation.

$$R = \frac{abc}{4A}.$$

We can use this property to compute the circumradius of a particular triangle, noting that area can be readily determined by Heron's Formula. As an exercise, attempt to find the circumradius of an equilateral triangle with side length x as well as the circumradius of a right triangle with side lengths a, b, and $\sqrt{a^2 + b^2}$. For the latter case, you will intriguingly find

that the circumradius is equal to half of the hypotenuse and that the circumcenter lies at the midpoint of the hypotenuse.

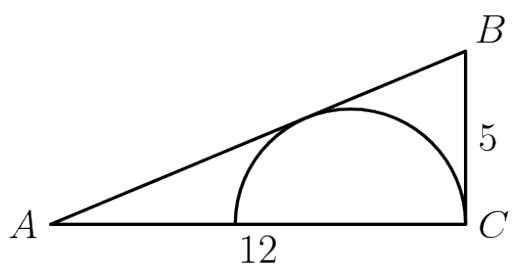
Moving onward, we consider the incircle of a particular triangle ΔABC, which is the largest circle that can be contained inside the triangle. The incircle is always tangent to all three sides, and the incenter (center of the incircle) is housed at the intersection of the triangle's angle bisectors.

The radius of the incircle r, known succintly as the *inradius*, is gift-wrapped in the elegant equation $A = sr$, where s is the semiperimeter and A is the area.

$$s = \frac{a+b+c}{2} = \frac{\text{perimeter}}{2}.$$

Problem Batch C

1. In the right triangle ΔABC, $AC = 12$, $BC = 5$, and angle C is a right angle. A semicircle is inscribed in the triangle as shown. What is the radius of the semicircle? (AMC 8 2017) *Hint: Reflect B across the line AC to form a larger triangle with a fully fledged incircle.*



2. Square ABCD has side length 2. A semicircle with diameter AB is constructed inside the square, and the tangent to the semicircle from C intersects side AD at E. What is the length of CE? (AMC 10A 2004)

3. Prove that the circumcenter of a right triangle lies at the midpoint of the hypotenuse. You can employ analytical geoemtry by assigning each vertex a coordinate in the xy plane and searching for the point equidistant to all three vertices.

4. A sphere with center O has radius 6. A triangle with sides of length $15, 15$, and 24 is situated in space so that each of its sides is tangent to the sphere. What is the distance between O and the plane determined by the triangle? (AMC 10A 2019)

5. In $\triangle ABC$ with side lengths $AB = 13$, $AC = 12$, and $BC = 5$, let O and I denote the circumcenter and incenter, respectively. A circle with center M is tangent to the legs AC and BC and to the circumcircle of $\triangle ABC$. What is the area of $\triangle MOI$? (AMC 12B 2018)

Law of Sines and Cosines

You can navigate through this section if you are familiar with trigonometry. Otherwise, we suggest browsing the website below to acquire a baseline understanding.

https://www.mathsisfun.com/algebra/trigonometry.html

As a note regarding notation, we set the side lengths $a = |BC|$, $b = |AC|$, and $c = |AB|$, as well as the angles $A = \angle BAC$, $B = \angle ABC$, and $C = \angle BCA$. We can connect the sines of the angles and the side lengths by employing the **Extended Law of Sines**.

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)} = 2R,$$

Here, R is the circumradius of ΔABC. A corollary of this principle is that we are aware of two angle measurements and one side length of a triangle, we can figure out all the other side lengths and angle measurements. We will use this Law to prove the area formula below.

$$\boxed{A = \frac{1}{2}ab\cos(C)}.$$

PROOF

We know that $R = \frac{abc}{4A}$ from earlier, which means that $A = \frac{abc}{4R}$ for circumradius R. According to the Law of Sines, $2R = \frac{c}{\sin(C)}$, which means that

$$A = \frac{abc}{2(2R)} = \frac{abc}{2 * \frac{c}{\sin(C)}} = \frac{1}{2}ab\sin(C).$$

This completes the proof.

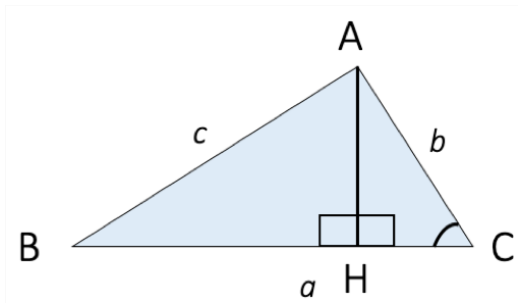
If we are endowed with two side lengths and the angle between them, we can also calculate all the other measurements. This will require the **Law of Cosines**, which states that

$$c^2 = a^2 + b^2 - 2ab\cos(C).$$

This equation has elegant symmetry in the sense that the similar equations $a^2 = b^2 + c^2 - 2bc\cos(A)$ and $b^2 = a^2 + c^2 - 2ac\cos(B)$ are also valid. We will sketch the proof by considering the case $\cos(C) > 0$ as all the other cases are similar and can be proved analagously.

PROOF

Consider the triangle ΔABC with side lengths $|AB| = c$, $|AC| = b$, and $|BC| = a$. We consider the altitude from vertex A, which intersects side BC orthogonally at point H. Note that we generate right triangles ABH and ACH.



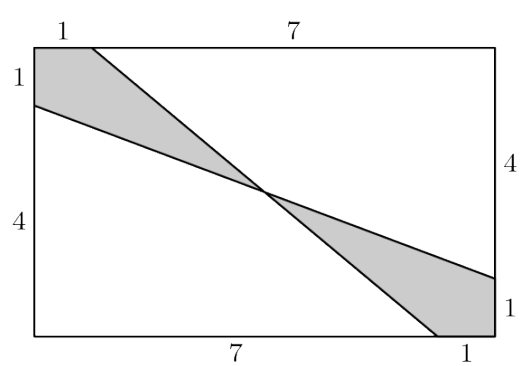
We consider the right triangle ΔACH with hypotenuse b and specifically the angle at vertex C. We have $\cos(C) = \frac{CH}{AC}$ and $\sin(C) = \frac{AH}{AC}$, which means that $CH = AC\cos(C) = b\cos(C)$ and $AH = AC\sin(C) = b\sin(C)$. Thus, $BH = BC - HC = a - b\cos(C)$. By Pythagorean Theorem on ΔABH, we have

$$\begin{aligned} AB^2 &= BH^2 + AH^2 \\ &= (a - b\cos(C))^2 + (b\sin(C))^2 \\ &= a^2 - 2ab\cos(C) + b^2\cos^2(C) + b^2\sin^2(C) \\ &= a^2 + b^2[\cos^2(C) + \sin^2(C)] - 2ab\cos(C) \\ &= a^2 + b^2 - 2ab\cos(C). \end{aligned}$$

Note that $\cos^2(C) + \sin^2(C) = 1$ by the Pythagorean Identity. Therefore, we have $c^2 = a^2 + b^2 - 2ab\cos(C)$, which is the verbatim statement of the Law of Cosines.

Problem Medley

1. Find the area of the shaded region.



2. Three similar triangles have areas of 4, 16, and 36. If the smallest triangle has a perimeter of $4 + 4\sqrt{2}$, what is the sum of the perimeters of the other two triangles?

3. Triangle ABC has a right angle at B. Point D is the foot of the altitude from B, $AD = 3$, and $DC = 4$. What is the area of $\triangle ABC$? (AMC 10A 2009)

4. A triangle with side lengths in the ratio $3 : 4 : 5$ is inscribed in a circle with radius 3. What is the area of the triangle? (AMC 12A 2007)

5. Points $A(11, 9)$ and $B(2, -3)$ are vertices of $\triangle ABC$ with $AB = AC$. The altitude from A meets the opposite side at $D(-1, 3)$. What are the coordinates of point C? (AMC 10B 2017)

6. All three vertices of $\triangle ABC$ are lying on the parabola defined by $y = x^2$, with A at the origin and $\overline{BC}$ parallel to the x-axis. The area of the triangle is 64. What is the length of BC? (AMC 10B 2016) *Hint: Note that the triangle is isosceles and find the length of its base and height supposing that the coordinates of B and C are $(-b, b^2)$ and (b, b^2), respectively.*

7. A thin piece of wood of uniform density in the shape of an equilateral triangle with side length 3 inches weighs 12 ounces. A second piece of the same type of wood, with the same thickness, also in the shape of an equilateral triangle, has side length of 5 inches. Which of the following is closest to the weight, in ounces, of the second piece? (AMC 10B 2016)

8. Joy has 30 thin rods, one each of every integer length from 1 cm through 30 cm. She places the rods with lengths 3 cm, 7 cm, and 15 cm on a table. She then wants to choose a fourth rod that she can put with these three to form a quadrilateral with positive area. How many of the remaining rods can she choose as the fourth rod? (AMC 10A 2017) *Hint: This is basically an extension of the triangle inequality. If $a \leq b \leq c \leq d$ are the side lengths of a quadrilateral, then $a + b + c > d$.*

9. In $\triangle ABC$ with a right angle at C, point D lies in the interior of $\overline{AB}$ and point E lies in the interior of $\overline{BC}$ so that $AC = CD$, $DE = EB$, and the ratio $AC : DE = 4 : 3$. What is the ratio $AD : DB$? (AMC 10B 2019)

10. Two different points, C and D, lie on the same side of line AB so that $\triangle ABC$ and $\triangle BAD$ are congruent with $AB = 9$, $BC = AD = 10$, and $CA = DB = 17$. The intersection of these two triangular regions has area $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$. (AIME II 2019)

11. An object moves 8 cm in a straight line from A to B, turns at an angle α, measured in radians and chosen at random from the interval $(0, \pi)$, and moves 5 cm in a straight line to C. What is the probability that $AC < 7$? (AMC 12A 2003)

12. A triangle has area 30, one side of length 10, and the median to that side of length 9. Let θ be the acute angle formed by that side and the median. What is $\sin(\theta)$? (AMC 12A 2012)

13. Prove the Pythagorean Theorem using the Law of Cosines. Recall that the cosine of a right angle is equal to 0.

14. In a given plane, points A and B are 10 units apart. How many points C are there in the plane such that the perimeter of $\triangle ABC$ is 50 units and the area of $\triangle ABC$ is 100 square units? (AMC 10B 2019)

15. In $\triangle ABC, AB = AC = 10$ and $BC = 12$. Point D lies strictly between A and B on $\overline{AB}$ and point E lies strictly between A and C on $\overline{AC}$ so that $AD = DE = EC$. Then AD can be expressed in the form $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$. (AIME I 2018)

16. A circle of radius 1 is randomly placed in a 15-by-36 rectangle ABCD so that the circle lies completely within the rectangle. Compute the probability that the circle will not touch diagonal AC. (AIME 2004)

Selected Problem Solutions

The two legs of a right triangle, which are altitudes, have lengths $2\sqrt{3}$ and 6. How long is the third altitude of the triangle? (AMC 10A 2014)

SOLUTION. We note that the hypotenuse of the triangle is equal to $c = \sqrt{(2\sqrt{3})^2 + (6)^2} = \sqrt{12 + 36} = 4\sqrt{3}$ by Pythagorean Theorem. Since the two legs are $2\sqrt{3}$ and 6, the area of the triangle is $\frac{1}{2} * 2\sqrt{3} * 6 = 6\sqrt{3}$. We consider the area formula $A = \frac{1}{2}(base)(height)$. The third altitude corresponds with the hypotenuse, which means that

$$\begin{aligned} \frac{1}{2}(4\sqrt{3}) * h &= 6\sqrt{3} \\ h &= \boxed{3}. \end{aligned}$$

Thus, the third altitude is equal to 3.

In $\triangle ABC$, $AB = 86$, and $AC = 97$. A circle with center A and radius AB intersects $\overline{BC}$ at points B and X. Moreover $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC? (AMC 10A 2013)

SOLUTION. It may help from the reader's point of view to draw a diagram while perusing this solution. We note that $AB = AX = 86$ since both X and B lie on the trajectory of the circle with center A. Let $BX = x$ and $CX = y$, which means that $BC = x + y$. Since $AC = 97$, using Stewart's Theorem, we have

$$\begin{aligned} BX * CX * BC + AX^2 * BC &= AB^2 * CX + AC^2 * BX \\ xy(x + y) + 86^2(x + y) &= 86^2 * y + 97^2 * x \\ x^2y + xy^2 + 86^2x &= 97^2x \end{aligned}$$

Since $x > 0$, then

$$\begin{aligned} xy + y^2 + 86^2 &= 97^2 \\ y(x+y) = 97^2 - 86^2 = (97+86)(97-86) = (183)(11) &= 3 * 61 * 11. \end{aligned}$$

Note that x and y are integers as stated in the problem. Since $x + y$ must be larger than y and $x + y = BC < AB + AC = 86 + 97 = 183$ by the Triangle Inequality, then we must select $y = 33$ and $x + y = 61$. Thus, BC is equal to $\boxed{61}$.

> A triangle has area 30, one side of length 10, and the median to that side of length 9. Let θ be the acute angle formed by that side and the median. What is $\sin(\theta)$? (AMC 12A 2012)

SOLUTION. Label the triangle ΔABC for the sake of clarity. Let the median described in the problem emanate from vertex A and intersect side BC at point M. Thus, $AM = 9$ and $BM = MC = 5$ as M is the midpoint of the side with length 10. Without the loss of generality, suppose that $\theta = \angle AMC$ is the acute angle and thus $\angle AMB = 180 - \theta$.

We individually consider the two sub-triangles ΔAMC and ΔAMB and note that their areas must add up to 30 as $[\Delta ABC] = 30$. We note $[\Delta AMC] = \frac{1}{2}(AM)(MC)\sin(\angle AMC)$ and $[\Delta AMB] = \frac{1}{2}(AM)(MB)\sin(\angle AMB)$ using the area formula on page 8. Thus,

$$\begin{aligned} [\Delta AMC] + [\Delta AMB] &= 30 \\ \frac{1}{2}(AM)(MC)\sin(\theta) + \frac{1}{2}(AM)(MB)\sin(180-\theta) &= 30 \\ \frac{1}{2}(9)(5)\sin(\theta) + \frac{1}{2}(9)(5)\sin(180-\theta) &= 30. \end{aligned}$$

We have $\sin(180 - \theta) = \sin(\theta)$ by trigonometric properties. This means that

$$\begin{aligned} \frac{1}{2}(9)(5)\sin(\theta) + \frac{1}{2}(9)(5)\sin(\theta) = 45\sin(\theta) &= 30 \\ \sin(\theta) &= \boxed{\frac{2}{3}}. \end{aligned}$$